CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

4.1.1 PC or LAPTOP

- Processor : Intel Core i5 / Core i7.
- Speed : 2.4 GHZ or above.
- RAM : Minimum 8 GB (Recommended 16 GB).
- Graphics : Dedicated GPU (Colab provides GPU for training)

4.2 SOFTWARE REQUIREMENTS

- Operating system : Windows 11 or any OS that supports a web browser
- Coding Language : Python 3.11+ and TypeScript 5.5
- IDE/Platform: Visual Studio Code
- Deep Learning Frameworks / Libraries :
    - Ultralytics YOLOv11s – PCB defect detection
    - Ultralytics YOLO11n – Electronic component detection
    - PyTorch 2.13.0, Torchvision 0.18+
- Supporting Librarires:
    - NumPy, OpenCV (image preprocessing & handling)
    - FastAPI, Express.js (backend services)
    - React, MongoDB (web interface & data storage)

4.3 SOFTWARE DESCRIPTION

- **PYTHON**

Python is a high-level programming language that is used extensively in artificial intelligence, machine learning, and computer vision-related applications. Due to its ease-of-use and the availability of a large number of libraries, it is a good choice for creating deep learning models. For this project, Python 3.11+ is being used for creating the PCB inspection pipeline, image pre-processing, defect detection, and decision support capabilities.